



FIRST SEMESTER 2026-27
COURSE HANDOUT

In addition to part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

COURSE NO. : MATH F421
COURSE TITLE : Combinatorial Mathematics
INSTRUCTOR IN CHARGE : KRISHNENDRA SHEKHAWAT (6067Q, Department of Mathematics)

COURSE DESCRIPTION: Combinatorics is often described briefly as being about counting, and indeed counting is a large part of combinatorics. As the name suggests, however, it is broader than this: it is about combining things. Questions that arise include counting problems:

“How many ways can these elements be combined?”

But there are other questions, such as whether a certain combination is possible, or what combination is the “best” in some sense.

This course presents the different methods to solve the counting problems. We will mainly focus on combinatorial approach instead of algebraic approach.

SCOPE and OBJECTIVES: Combinatorics is a fascinating and a broad subject. It’s hard to classify Combinatorics, but a common theme is that it deals with structures that are, in some sense, finite or discrete. What sets Combinatorics apart from other branches of Mathematics is that it focuses on *techniques* rather than results. The way a Theorem is proved is of bigger importance than the statement of the Theorem itself. Combinatorial reasoning underlies all analysis of Computer Systems. It plays a similar role in Discrete Operation Research Problems and in Finite Probability. It is very much useful in the analysis of the speed and logical structure of Operation Research Algorithms to optimize efficient manufacturing or garbage collections. It has been proven an efficient tool for Graph Counting.

COURSE LEARNING OUTCOMES: The course focuses on different techniques related to counting problems. At the end of the course, student will be aware of many existing methods used to solve counting problems and when these methods are applicable.

TEXT BOOK:

T1: Alan Tucker, Applied Combinatorics, John Wiley & Sons, 6th Edition, 2012.

T2: Allenby and Slomson, How to Count, CRC Press, 2nd Edition, 2011.

REFERENCE BOOKS:

R1. W.D. Wallis and J.C. George, Introduction to Combinatorics, CRC Press, 2th Edition, 2017.



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R2. Richard A. Brualdi, Introductory Combinatorics, Pearson, 4th Edition 2008.

R3. Miklos Bona, Introduction to Enumerative Combinatorics, Tata McGraw-Hill, 2007.

COURSE PLAN:

Week No.	Module No.	Description	Reference (Sections)	Learning outcomes
1	1. General Counting Methods for Arrangements and Selections	L1 Introduction to Combinatorics and Two Basic Counting Principles	Section 5.1 (T1)	Familiarize with Combinatorics and Understanding Counting Principles
		L2-L3 Arrangements, Selections and Distributions (with and without repetition)	Sections 5.2-5.4(T1)	
		L4 Binomial Identities	Section 5.5(T1)	
2-3	2. Generating Functions	L5-L7 Ordinary and Exponential Generating Functions	Sections 6.1-6.2, 6.4 (T1)	Introducing Generating Functions as a Mathematical Model for Counting Problems
		L8-L9 Partitions and Dot Diagrams	Chapter 6 (T2)	
		L10-L11 Summation (Product and composition of generating functions)	Section 6.5 (T1)	
4-5	3. Recurrence Relations	L12-13 Linear Recurrence Relation and their solutions	Sections 7.1-7.5 (T1)	Introducing Recurrence relations as a simplest way to solve Counting Problems
		L14-15 Non-linear Recurrence Relations	Section 7.8(T2)	
		L16 Recurrence Relations and Generating Functions		
6-8	4. Inclusion–Exclusion	L17- L18 Inclusion–Exclusion Formula	Sections 8.1-8.2(T1)	Introducing Inclusion–Exclusion Formula for solving more complex counting problems
		L19- L20 Coloring and	Section 2.4 (T1)	



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		Chromatic Polynomial		
		L21 Derangements	Section 4.2 (T2)	
		L22-L23 Rook Polynomials	Section 8.3 (T1)	
		L24-L25 Matching and Marriages	Section 17.2 (T2)	
9-10	5. Polya's Enumeration Formula	L26-L27 Burnside's Theorem	Sections 9.1-9.2(T1)	Introducing the relation between algebra and combinatorics
		L28 Polya's Formula	Sections 9.3-9.4(T1)	
		L29-L30 Counting Simple graphs	Section 14.6 (T2)	
11-12	6. The Pigeonhole Principle and Ramsey's Theorem	L31-L32 The Pigeonhole Principle and its Applications	Sections 15.1-15.3 (T2)	Introducing the relation between graph theory and combinatorics
		L33-L35 Ramsey's Theorem and Ramsey Numbers	Sections 16.1-16.2 (T2)	
13-14	7. Catalan and Stirling Numbers	L36-L38 Stirling Numbers of the Second kind and of the First kind	Sections 5.1-5.2 (T2)	Introducing numbers useful for enumeration
		L39-L40 Catalan Numbers	Sections 5.3 (T2)	

EVALUATION SCHEME:

Component	Duration (minutes)	Weightage	Date & Time	Nature of component (Close Book/ Open Book)



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Mid-semester Exam	90	35%		Close Book
Announced Quizzes*	20 each	20%	To be Announced Later	Closed Book
Comprehensive Exam	180	45%		Open Book

***Best two out of three quizzes will be considered.**

NC Criteria: Below 20 marks out of 200 or completely missed any one of the exam components

Attendance policy: Each student is expected to attend all classes and participate in all evaluation components.

CHAMBER CONSULTATION HOUR: To be announced in the class.

NOTICES: NALANDA web-site would be used to post course material as well as notices.

Makeup: Prior permission is needed for makeup; makeup will only be given if sufficient evidence is available for not being able to appear for the exams.

Krishnendra Shekhawat
Instructor-in-charge
Course No. MATH F421